Response to the Editor and Reviewers

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From Reactive to Predictive: A Federated Graph-Attention SAC Framework for Schedule-Driven WiFi Optimization in High-Density Transit Venues
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Dear Dr Botero,

thank you for the opportunity to revise our manuscript and thank you and the reviewers for the detailed and constructive comments. We have revised the paper substantially. This letter lists, point by point, how each comment was addressed, describes the additional experiments that were carried out, and states where we did not do what was suggested and why. Page and section numbers refer to the revised manuscript.

0. Summary of the revision, and an important correction to the reported results

Methodology and presentation (Reviewer 1). The method section was rewritten. It now defines the entities and their roles (per-AP agents, zone controllers, venue controller), the data each of them sees (Table 2 and Table 3), every term of the reward (with the weights' rationale and a sensitivity study), the constraint, the schedule features, and the privacy scope of the federation. A notation of what is learned where is given by the roles subsection, the dimension table (Table 4) and Algorithm 1.

Simulator realism (Reviewer 2). The simulator was extended with (i) MAC-layer contention based on Bianchi's 802.11 model, (ii) a channel-switching cost, and (iii) an EDCA-style quantised airtime mapping for the QoS queues. All methods are re-evaluated in the extended simulator, and robustness experiments quantify the effect of each assumption.

Correction of the reported results (Editor: "ensure the results are accurately reported"). While preparing the revision we audited the code and the way the results of the first submission had been produced, and we found problems that we must report to you, because they invalidate the numbers of the original Tables 2–3 and Figures 2–6:

- The numbers in the original results tables and in the convergence, scalability and ablation figures were not generated by the released code: the plotting script contained them as constants (and the convergence curves were synthetic curves of the form $f(1 - e^{-t/\tau})$ plus noise). We have discarded all of them. **Every number, table and figure of the revised paper is now computed from logged runs** by a script in the repository (`analysis/aggregate.py`); the raw per-run records are part of the repository.
- The LSTM-DRL and FedDDPG baselines had been evaluated with randomly initialised, untrained networks. They are now trained with the same protocol as the proposed method.
- In ablation A2 ("no schedule") the schedule still entered the agent state through a second path; in A4 ("no hierarchical federation") federation was switched off completely rather than only its global level, and in A1 ("no joint gradient") the predictor was not trained at all. These ablations were re-implemented as described in the text.
- The capacity constraint was imposed on the number of users attached to an AP, a quantity no action can change, so the Lagrange multiplier could never act. The constraint is now on airtime utilisation, which the actions can influence.
- The communication saving of "11.5×" was asserted, not measured; in the implemented scheme

weights were exchanged at both levels. The global level now exchanges soft policy outputs on a public reference set and all traffic is counted byte by byte.

- The statistics claimed in the original (Wilcoxon tests with Bonferroni correction over 5 seeds) cannot reach $p < 0.05$ with 5 seeds, and no test code existed. We now use paired bootstrap confidence intervals and exact sign-flip tests, with the number of seeds stated.
- The WACA, Dartmouth and LCR-HDD datasets were not used by any experiment (the original text claimed that WACA calibrated the interference coefficients). The revised text states that the simulator is not calibrated to measurements and that no measured traces are used.
- The original text stated that pre-trained checkpoints were released; no checkpoints were available and the claim has been removed.

Consequence for the conclusions. Under the corrected and more demanding evaluation the headline results of the first submission (reward 0.763, latency 32.4 ms, +24.5% over FedDDPG) are not reproduced, and we have rewritten the abstract, the discussion and the conclusions accordingly (see Section 1 of this letter and the revised Sections 4–6). In short: the corrected evidence supports the *forecasting* contribution (timetable-conditioned forecasts reduce the RMSE by 32% relative to persistence and by 27% relative to the same predictor without the schedule, 31% in surge windows; at the one-step horizon simple forecasters are better), shows that the learned federated SAC agents are stable and clearly better than learned DDPG baselines (reward 0.141 against 0.116 and 0.108), but it does *not* show that they outperform the static incumbent configuration (SSF, 0.181). Two of the original contributions, feeding the forecaster's embedding to the policy and the joint predictor–policy gradient, did not show a benefit and are reported as negative results (ablations A3, A1 and an η sweep). We understand that this weakens the paper's claims; we think it is the only defensible position given the evidence.

1. Reply to the Editor

"Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained."

Response. See Section 0 above. All results were regenerated; the abstract, introduction, discussion and conclusion no longer contain the superlatives and absolute novelty statements of the first version ("first", "none of the above works", "no prior work" were replaced by statements about the works we reviewed); a dedicated subsection *Discussion and Limitations* (Section 5) lists what is and is not supported by the evidence. The code release statement was corrected.

2. Reviewer 1

R1.1 – "The presentation of the idea from a methodological point of view is unclear. Many concepts are loosely presented, as if some information would be missing."

Response. We rewrote Section 3 and added what was missing: (i) Section 3.1 defines the three kinds of entities, who observes what, and the time and data flow of one control interval; (ii) Tables 2 and 3 list every input of the framework with its source and dimension, including the eight schedule features; (iii) Section 3.2 defines the action, each reward term with its normalisation, and the constraint; (iv) Table 4 and Eq. (3) give all layer dimensions of ST-GCAT; (v) Section 3.4 specifies the actor, the critic and the update equations, and Algorithm 1 was rewritten so that it matches

the implementation step by step. A table of the hyperparameters of all methods (Table 6) was added.

R1.2 – "Some concepts, for instance 'per-AP agents', do not appear described."

Response. Section 3.1 now defines the per-AP agent: an independent SAC agent for each AP, consisting of an actor with a federated base and a personal head, twin critics with target copies and a private replay buffer; it runs on the AP or on its zone controller, observes only the quantities of Table 2, and chooses power, channel and per-class airtime of *its own* AP every 5 minutes. Section 3.4 gives the architecture and updates, and Section 3.6 states which of its parameters are federated (the 270,080 parameters of the base layers; heads and critics stay local).

R1.3 – "Many of the parameters are ad-hoc, without a detailed description on their derivation (see for instance the numbers for the multi-dimensional reward function in Section 3.1)."

Response. The reviewer is right that these weights are not derived, and we now say so explicitly instead of writing that they are "tuned per the design specification". Section 3.2 (i) defines every reward term on a unit scale (reference throughput 100 Mbps and reference latency 100 ms for the normalisation), (ii) states that the five weights sum to one and encode a priority order (throughput $>$ latency $>$ fairness; the QoS term is small because violations already reduce the throughput and latency terms; the prediction term is a small auxiliary signal), and (iii) states that they are a design choice, not an optimum. To show how much the conclusions depend on them we added a sensitivity study (Section 4.9, Table 13) with three alternative weightings (equal weights, throughput-heavy, latency-heavy). We also made the other previously implicit design constants explicit in Table 6 (constraint limit, multiplier step and bound, reward scale, embedding scale, number of distillation steps, etc.) and state how the one tuned value (the embedding scale κ) was chosen. We did not tune any hyperparameter separately per method; we must add that the shared design choices made during development (reward scale, initial entropy temperature, update-to-data ratio, initialisation at the incumbent configuration, embedding scale) were selected on validation episodes of one seed, mostly using variants of the proposed method, which if anything favours it over the baselines; this is stated in Section 5.

R1.4 – "Another aspect which is unclear is how the historical information is presented to the framework."

Response. Section 3.1 (time and data flow) and Table 2 specify it: every AP reports five counters every interval (normalised user load, throughput, mean latency, airtime utilisation, user count); the venue controller keeps the last $T = 12$ intervals (one hour) of all APs as the tensor $\mathbf{H} \in \mathbb{R}^{N\times 12\times 5}$ (the oldest frame is repeated at the start of an episode), together with the interference graph $\mathbf{A}$ and the schedule tensor $\mathbf{S} \in \mathbb{R}^{N\times 6\times 8}$ built from the timetable of the next 30 minutes (Table 3). The shapes of all projections are in Table 4, and Section 3.3 states how the forecast targets are built (true future loads, stored once observed, with masking at the end of an episode).

R1.5 – "Why federation is actually required. The aspects of privacy involved are not completely clear in order to justify this architectural decision."

Response. We rewrote the justification and limited its scope (Section 3.1, "What is federated, and why"). The argument is that the per-user data used to train the AP agents (association records, device identifiers, traffic mix, mobility) cannot always be pooled when the zones of a venue are operated by different parties; federation lets agents share what they learned while transitions, per-user records and replay buffers stay at the AP. The predictor uses only AP-level aggregates that any WLAN controller already collects. We state that this is data minimisation and *not* a formal privacy guarantee (no differential privacy or secure aggregation; parameters and outputs can leak

information) and list this in the limitations. Because the benefit of federation was previously only asserted, we added an experiment (Section 4.7, Table 11) that compares the same agents with no federation, flat FedAvg, zone-level FedAvg only, weight-based hierarchical federation, the proposed distillation-based variant, and a (privacy-violating) learner that pools all experience, and that reports measured communication. Result: federated agents are better than isolated ones (reward 0.141 against 0.121); inter-zone distillation needs 34× less wide-area traffic than weight-based hierarchical federation and 1711× less than flat FedAvg at a similar reward; pooling all experience did not help (0.100). The reward differences among the federated variants are within seed noise and we do not rank them.

R1.6 – "The reviewer suggests the author to present a more clear and detailed description, avoiding the assumption that many aspects are clear in advance for the reader."

Response. In addition to the points above, the introduction now states the setting and the four gaps in plain terms and points to the system model; every symbol is defined where first used; the simulator, the baselines and the evaluation protocol are described in a separate, self-contained set of subsections (4.1–4.3), including the exact definition of every metric.

3. Reviewer 2

R2.1 – MAC-layer modelling: "The authors must explicitly quantify or discuss how the omission of MAC-layer contention affects the policy's performance metrics, particularly latency, which is reported as a clean 32.4 ms."

Response. We agree, and we did not only discuss it: we modelled it and measured the effect. The simulator now includes (Section 4.1) carrier-sense domains in which co-channel APs time-share the medium, the saturation efficiency of the 802.11 DCF obtained from Bianchi's model for the number of contending stations ($\eta = 0.87$ for 1 station, 0.70 for 10, 0.50 for 100), and an M/G/1-type queueing delay built on the domain service time, so latency is no longer a constant but depends on contention, load and the airtime budget of each class. Quantification (Section 4.8, Table 12): the same policies evaluated without contention obtain a latency of 92.5 ms instead of 170.6 ms (STAG-FedSAC) and 70.3 ms instead of 124.6 ms (SSF), i.e. ignoring contention understates latency by roughly a factor of 1.8. We also trained policies in the contention-free simulator and evaluated them with contention; the table shows what this "contention-unaware planning" costs. The headline figure of 32.4 ms of the first submission is withdrawn. A residual limitation is that hidden terminals, adjacent-channel leakage and frame-level dynamics are not modelled; this is stated in Section 5.

R2.2 – Expenses of channel switching: "The authors should account this penalty."

Response. The simulator now charges every channel change an outage of $t_{\mathrm{sw}} = 3$ s of the 300 s interval (channel-switch announcement plus re-association of clients that miss it): the AP's users lose that fraction of service and the packets queued during the outage see on average half of it as extra delay; this enters throughput, latency, reward, QoS and the switch count (Section 3.7). The agents are trained with this cost. Section 4.8 reports the results for $t_{\mathrm{sw}} \in \{0, 3, 10\}$ s and for a policy trained without any switching cost and evaluated with it; the average number of channel switches per AP and interval of each method is part of Table 7. The value of t_{sw} is an engineering assumption and not a measurement; this is stated explicitly.

R2.3 – Bandwidth mapping to QoS queues: "The authors need to bridge this architectural gap."

Response. Section 3.7 now bridges it. VoIP, video and best-effort are mapped to the AC_VO, AC_VI and AC_BE access categories of 802.11e EDCA. Since EDCA offers relative priority rather than a bandwidth split, the simplex action is interpreted as an *airtime budget per access category*, enforced on top of EDCA with the per-AC airtime quota scheduler available in current AP software, with EDCA parameters left at their standard values. To make the action implementable we quantise it to multiples of 1/16 with a minimum of 1/16 per category and treat the budgets as work-conserving limits (unused airtime is redistributed). The simulator applies exactly this quantisation and redistribution to every method, including the baselines. What we could not do is validate the mapping on hardware, and the fourth (background) category and vendor differences in exposing airtime quotas are not evaluated; both are stated in Section 5.

R2.4 [nit] – "In Section 3.2.1, define the exact dimensions of the projection matrices and the attention vector a to complement the architectural data provided in Table 1."

Response. Done: Table 4 lists all learned maps of ST-GCAT, and Eq. (3) now defines the attention explicitly: $\mathbf{W}^{\mathrm{sp}} \in \mathbb{R}^{128\times128}$ (output split into $K = 4$ blocks of $d_k = 32$), one attention vector $\mathbf{a}^k = [\mathbf{a}_l^k; \mathbf{a}_r^k] \in \mathbb{R}^{64}$ per head, input projections 5×128 and 8×128, and the equivalent decomposition used in the implementation. While doing this we noticed that the implementation used a single attention vector shared by all heads; it now uses one per head, as the equations say.

4. Additional experiments carried out

All experiments use the extended simulator. Each learned method is trained for 1000 episodes (200 warm-up) with three seeds for the main comparison (the first version stated 2000 episodes; the learning curves flatten earlier and every method had to be re-run); the other experiments use one to three seeds as stated in each table; evaluation uses 20 held-out episodes with seeds shared by all methods.

- Main comparison and ablations (Tables 7, 9) with paired confidence intervals.
- Federation study (Table 11) with measured communication.
- Robustness to MAC contention and to the channel-switching cost (Table 12).
- Reward-weight sensitivity (Table 13).

5. Points on which we deviated from what might be expected

- We did not replace the Python simulator with a packet-level simulator (e.g. ns-3); this would be a substantial separate study. We instead added the MAC-level effects the reviewer asked about, quantified their influence, and list the remaining ones as limitations.
- We did not validate against measured traces, and we removed the statement that such traces were used.
- The reward weights were not "derived"; we report that they are design choices and how sensitive the results are to them.

We thank the reviewers again; their comments led to a more careful and, we believe, more honest paper.

Sincerely,
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